

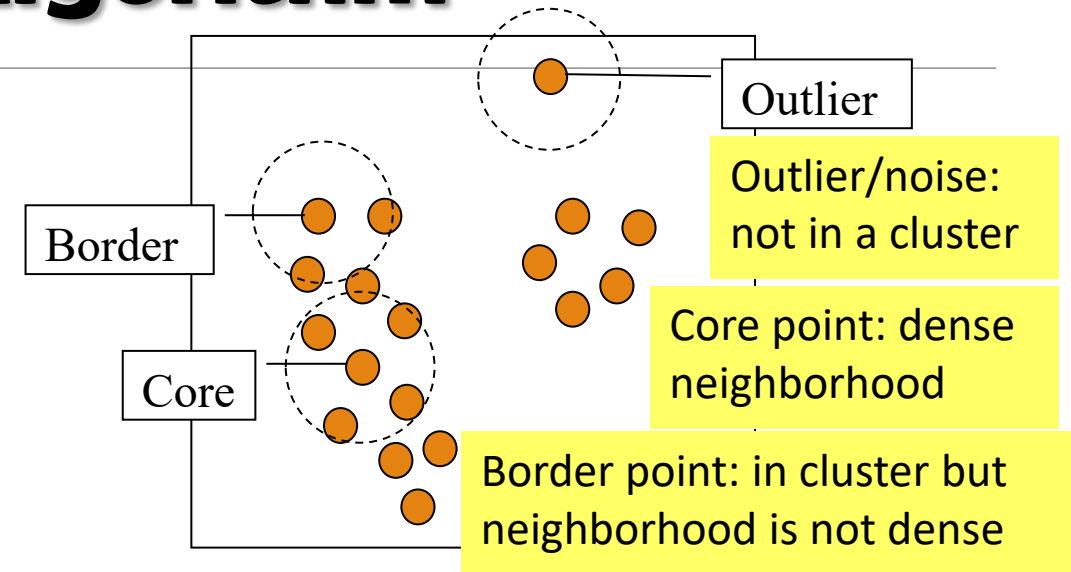
DBSCAN: The Algorithm

Algorithm

- Arbitrarily select a point p
- Retrieve all points density-reachable
 - from p w.r.t. Eps and $MinPts$
- If p is a core point, a cluster is formed
- If p is a border point, no points are directly density-reachable from p , and DBSCAN visits the next point of the database
- Continue the process until all of the points have been processed

Computational complexity

- If a spatial index is used, the computational complexity of DBSCAN is $O(n \log n)$, where n is the number of database objects
- Otherwise, the complexity is $O(n^2)$



DBSCAN Is Sensitive to the Setting of Parameters

Figure 8. DBScan results for DS1 with MinPts at 4 and Eps at (a) 0.5 and (b) 0.4.

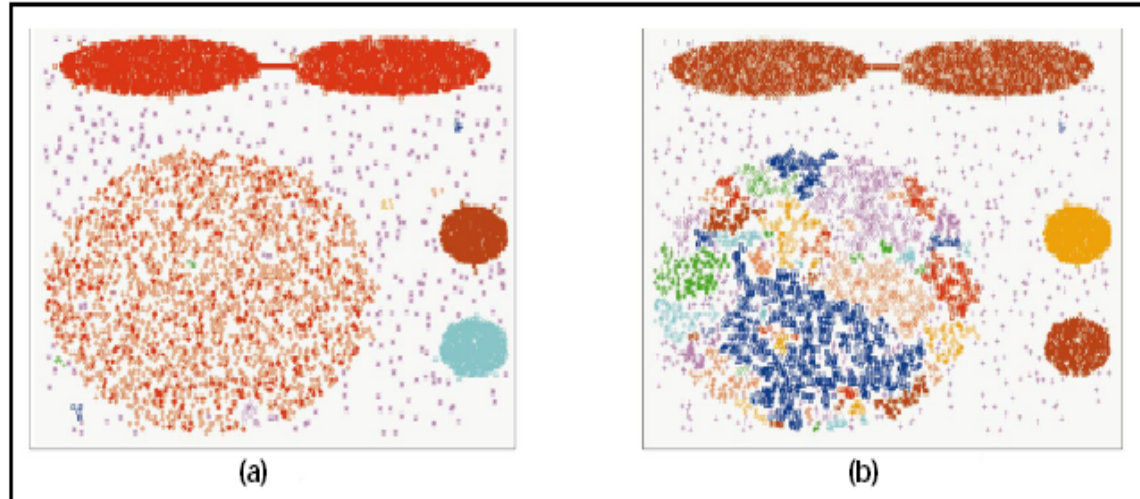
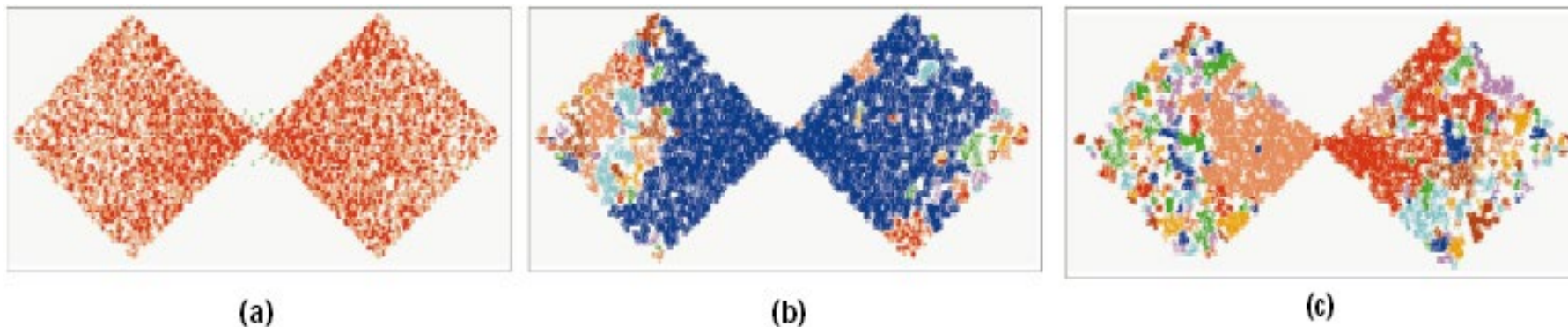


Figure 9. DBScan results for DS2 with MinPts at 4 and Eps at (a) 5.0, (b) 3.5, and (c) 3.0.



Ack. Figures from G. Karypis, E.-H. Han, and V. Kumar, *COMPUTER*, 32(8), 1999